HAT-P-20b

R-1339

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-20_190418` · created 2026-07-31T21:27:34+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458591.6891 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.017
Transit depth (Rp/Rs)^2	1.73 +/- 0.46 [%]
Orbital Inclination (inc)	87.6 +/- 1.2
Airmass coefficient 1 (a1)	1.238 +/- 0.013
Airmass coefficient 2 (a2)	-0.1124 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	None
Optimal Aperture	4.8
Optimal Annulus	7.91
Transit Duration (day)	0.082 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.017 vs 0.1284 (deviation 0.2σ)
Duration fit vs published	0.082 d vs 0.0775 d (deviation 0.35σ)
Mid-time O–C	-1.9 min
Depth SNR	7.2
Residual scatter	1.07 %

Light curve

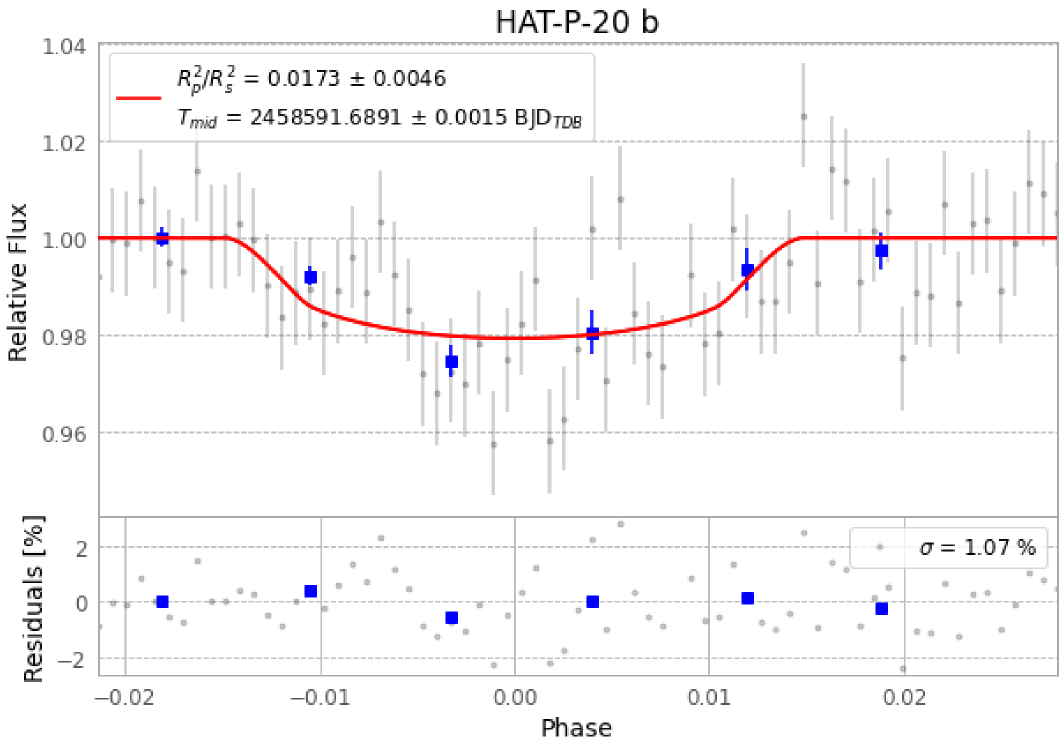

Figure 1 — FinalLightCurve_HAT-P-20 b_2019-04-17.png (SHA-256 b61685879f9eac59…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [304, 221], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 57a57c36ec5ddade…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

69 FITS frames (45 MB), `HATP-20190418030312.FITS` → `HATP-20190418062712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-190418.log (9b1ca0c34e53…), FinalLightCurve_HAT-P-20 b_2019-04-17.png (b61685879f9e…), FinalLightCurve_HAT-P-20 b_2019-04-17.csv (b031a988bbe9…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)